

Algebra and Discrete Mathematics (ADM)

Tutorial 6 Matrix operators

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Matrix transformations

- Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ with standard matrix

$$[T] = \begin{pmatrix} -1 & -1 \\ 2 & 3 \\ 3 & 1 \end{pmatrix}$$

- Find the image of the vector $\mathbf{x} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$

$$T(\mathbf{x}) = \begin{pmatrix} -1 & -1 \\ 2 & 3 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -5 \\ 14 \\ 7 \end{pmatrix}$$

Matrix transformations

- Consider $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with standard matrix

$$[T] = \begin{pmatrix} -2 & -1 & 0 \\ 1 & 2 & 3 \end{pmatrix}$$

- Find the image of the vector $\mathbf{x} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$

$$T(\mathbf{x}) = \begin{pmatrix} -2 & -1 & 0 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} -11 \\ 16 \end{pmatrix}$$

Reflection operators on $\mathbb{R}^2$

- Reflection about the x -axis, $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
- Reflection about the y -axis, $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
- Reflection about the line $y = x$, $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
- Reflection about the line $y = \sqrt{3}x$
 - $y = \sqrt{3}x$ makes an angle $\pi/3$ ($= 60^\circ$) with positive x -axis

$$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

Reflection operators on $\mathbb{R}^3$

- Reflection about the xy -plane

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

- Reflection about the xz -plane

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Reflection about the yz -plane

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Projection operators on $\mathbb{R}^2$

- Orthogonal projection onto the x -axis

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

- Orthogonal projection onto the y -axis

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Projection operators on $\mathbb{R}^3$

- Orthogonal projection onto the xy -plane

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

- Orthogonal projection onto the xz -plane

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Orthogonal projection onto the yz -plane

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Rotation operators on $\mathbb{R}^2$

- Moves points *counterclockwise* about the origin through a positive angle θ
- *Rotation matrix*

$$R_\theta := \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

- Clockwise about the origin through an angle θ

$$R_{-\theta} := \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

Rotation operators on $\mathbb{R}^3$

Operator	Rotation equations	Standard matrix
Counterclockwise rotation about the positive x -axis through an angle θ	$\begin{aligned} w_1 &= x \\ w_2 &= y \cos \theta - z \sin \theta \\ w_3 &= y \sin \theta + z \cos \theta \end{aligned}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$
Counterclockwise rotation about the positive y -axis through an angle θ	$\begin{aligned} w_1 &= x \cos \theta + z \sin \theta \\ w_2 &= y \\ w_3 &= -x \sin \theta + z \cos \theta \end{aligned}$	$\begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$
Counterclockwise rotation about the positive z -axis through an angle θ	$\begin{aligned} w_1 &= x \cos \theta - y \sin \theta \\ w_2 &= x \sin \theta + y \cos \theta \\ w_3 &= z \end{aligned}$	$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Dilations and contractions

- For $\alpha \geq 0$, dilation/contraction with factor α on $\mathbb{R}^2$, $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha x \\ \alpha y \end{pmatrix}$

$$\begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}$$

- For $\alpha \geq 0$, dilation/contraction with factor α on $\mathbb{R}^3$, $T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \alpha x \\ \alpha y \\ \alpha z \end{pmatrix}$

$$\begin{pmatrix} \alpha & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha \end{pmatrix}$$

Expansions and compressions on $\mathbb{R}^2$

- In the x -direction – $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha x \\ y \end{pmatrix}$

$$\begin{pmatrix} \alpha & 0 \\ 0 & 1 \end{pmatrix}$$

- In the y -direction – $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ \alpha y \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 \\ 0 & \alpha \end{pmatrix}$$

Shears on $\mathbb{R}^2$

- Shear in the x -direction by a factor α , $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + \alpha y \\ y \end{pmatrix}$

$$\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$$

- Shear in the y -direction by a factor α , $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y + \alpha x \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix}$$

Composition of matrix transformations

- Consider a square $ABCD$ with vertices

$$A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad C = \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \quad D = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

- Perform the following transformations
 - T_1 : shear in the x -direction by a factor 2
 - T_2 : reflection about the x -axis
 - T_3 : reflection about the y -axis

$$[T_1] = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad [T_2] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad [T_3] = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$[(T_3 \circ T_2 \circ T_1)] \begin{pmatrix} 1 & 3 & 3 & 1 \\ 1 & 1 & 3 & 3 \end{pmatrix} = [T_3][T_2][T_1] \begin{pmatrix} 1 & 3 & 3 & 1 \\ 1 & 1 & 3 & 3 \end{pmatrix} = \begin{pmatrix} -3 & -5 & -9 & -7 \\ -1 & -1 & -3 & -3 \end{pmatrix}$$